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Environmental Product Declaration

EPD INTERNATIONAL EPD SYSTEM

INDIA EPD INTERNATIONAL EPD SYSTEM

ECO PLATFORM EPD VERIFIED

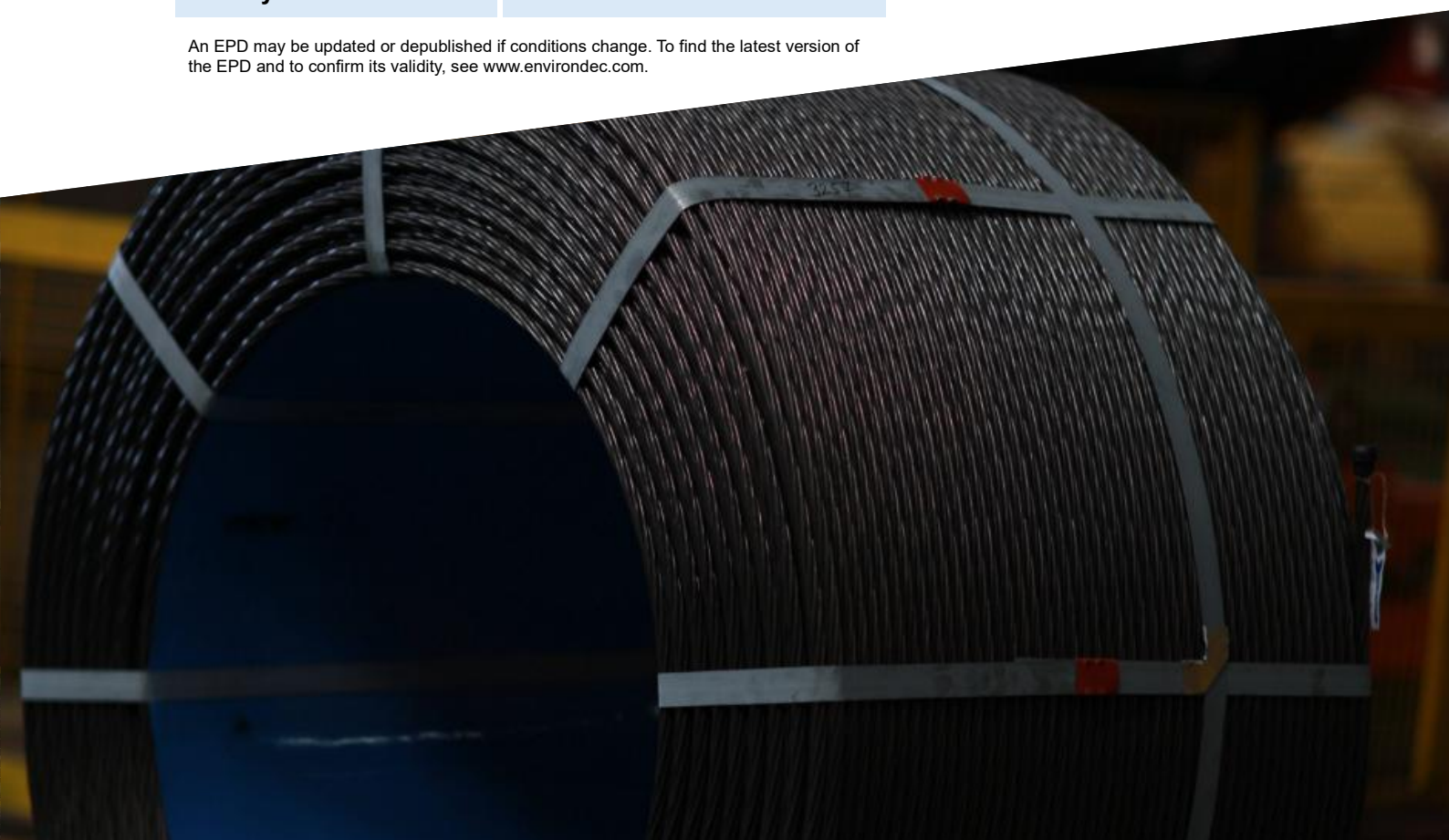
In accordance with ISO 14025 and EN15804:2012+A2:2019/AC:2021 for

Low Relaxed Pre-stressed concrete (LRPC) wire

from Tata Steel Limited

Programme:	The International EPD® System, www.environdec.com
Programme operator:	EPD International AB
EPD Owner	Tata Steel Limited
Licensee	EPD India
Type of EPD	EPD of single product
EPD registration number:	EPD-IES-0021821
Version date:	2025-11-14
Validity date:	2030-11-13

An EPD may be updated or depublished if conditions change. To find the latest version of the EPD and to confirm its validity, see www.environdec.com.



General Information

Programme Information		
Programme:	The International EPD® System	Licensee: EPD India
Address:	EPD International AB Box 210 60, SE-100 31 Stockholm Sweden	EPD India, a licensee of the International EPD® System 422, Midas, Sahar Plaza Mumbai, India-400059
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Accountabilities for PCR, LCA and independent, third-party verification	
Product Category Rules (PCR)	
CEN standard EN 15804 serves as the Core Product Category Rules (PCR)	
Product Category Rules (PCR):	*PCR 2019:14 Construction products, version 2.0.1 valid until 2030.04.07. UN CPC code 41263*
PCR review was conducted by:	*The Technical Committee of the International EPD System. A full list of members is available on www.environdec.com. Review chair: Rob Rouwette (chair), Noa Meron (cochair). The review panel may be contacted via the Secretariat support@environdec.com*
Third-party verification	
Independent third-party verification of the declaration and data, according to ISO 14025:2006, via	☒ EPD verification by individual verifier
Third-party verifier:	*Dr. Hüdai Kara,* *Metsims Sustainability Consulting,* *4 Clear Water Place, Oxford OX2 7NL, UK* *Email: hudai.kara@metsims.com*
Approved by:	The International EPD® System
Procedure for follow-up of data during EPD validity involves third party verifier: ☐ Yes ☒ No	

The EPD owner has the sole ownership, liability, and responsibility for the EPD. EPDs within the same product category but registered in different EPD programmes, may not be comparable. For two EPDs to be comparable, they must be based on the same PCR (including the same version number) or be based on fully-aligned PCRs or versions of PCRs; cover products with identical functions, technical performances and use (e.g. identical declared/functional units); have equivalent system boundaries and descriptions of data; apply equivalent data quality requirements, methods of data collection, and allocation methods; apply identical cut-off rules and impact assessment methods (including the same version of characterisation factors); have equivalent content declarations; and be valid at the time of comparison. For further information about comparability, see EN 15804 and ISO 14025

Company Information

Owner of the EPD:	Tata Steel Limited
Contact	Amit Agarwal
E-mail:	amit.agarwal1@tatasteel.com
LCA Practitioner	Tata Steel Limited

Description of the organization

Tata Steel Limited a flagship company of Tata Group is a multinational steel-making company headquartered in Mumbai (Maharashtra, India). The company is one of the world's most geographically diversified steel producing companies. It was established in India as Asia's first integrated private steel company in 1907 and today the company, together with its subsidiaries, associates and joint ventures, has its presence across five continents with key operations in India, the Netherlands and the United Kingdom. Tata Steel is one of the most diversified integrated steel producers in the world, with an annual crude steel production capacity of 35 MTPA. The Company's manufacturing assets are spread across India, the Netherlands, the UK, and Thailand. In India, Tata Steel operates in Jamshedpur and Gamharia in Jharkhand, as well as in Kalinganagar and Meramandali in Odisha, with an overall capacity of 21.6 MTPA. Additionally, the Company has downstream manufacturing facilities for wires, tubes, bearings, agricultural equipment, pig iron, ductile iron (DI) pipes, and various industrial by-products. The Company's operations also include Tata Growth Shop, a heavy-duty engineering and fabrication business, along with a Ferro Alloys and Minerals Division. The company's largest steel plant is located in Jamshedpur, Jharkhand (India). The company's operations in India are fully integrated from mining to finished steel production and it also has captive iron ore and coking coal mines in India.

Product-related or management system-related certifications
All facilities at Tata Steel Limited are ISO 9001, ISO 14001 & ISO 45001 certified.

Name and location of production site(s)	
Name	**Location**
Tata Steel Limited	Tarapur, Maharashtra, India

Product Information

Product name	7 wire Low Relaxation Strand for pre-stressing of concrete (LRPC strand)
Product identification	LRPC Strand is supplied in coil form with typical weight of 3 to 5 tons, covered with plastic sheets or other suitable material. A label indicating size, standard specification, tensile grade, name of the manufacturer etc. is attached to the coil
Product description	7 wire low relaxation strand for prestressing of concrete (LRPC Strand) is comprised of 6 high tensile wires wrapped around a central wire. It is thermo- mechanically processed to ensure the wires do not unravel. It is used in construction of concrete structures where pre-stressing is required to counter tensile loads acting during the lifetime – such as girders for road, river & railway bridges and flyovers, domes of atomic reactor, silos,

	hangars, aqueducts, LNG terminals, slabs of high-rise buildings, rock anchoring, windmills towers, viaducts and railway sleepers.
UN CPC code	41263
Geographical scope	India
Name of manufacturer	Tata Steel Limited
Location of production site(s)	Tarapur (Maharashtra), Pithampur (Madhya Pradesh)

Manufacturing process

Manufacturing of LRPC strand involves drawing, stranding and stress relieving operations. Wire rods are first subjected to surface preparation (pickling), followed by cold drawing to specific size as per the diameter of strand. 7 such wires- one core and 6 outer- are stranded together to achieve specific lay length. The strand is then subjected to thermomechanical treatment of simultaneous plastic stretching and heating to achieve compactness and low relaxation property. It is layer wound in coil form and covered with sheets for rust protection.

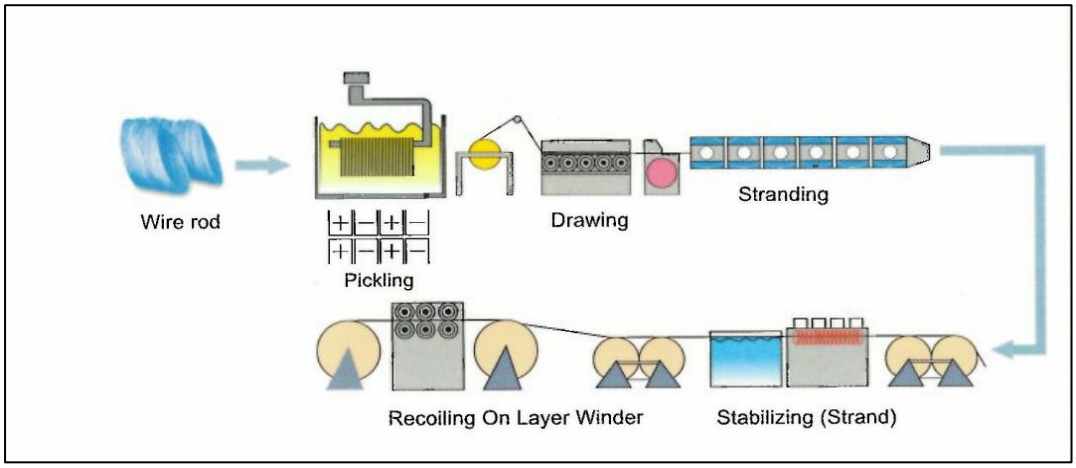

Figure: Process overview for LRPC wire manufacturing

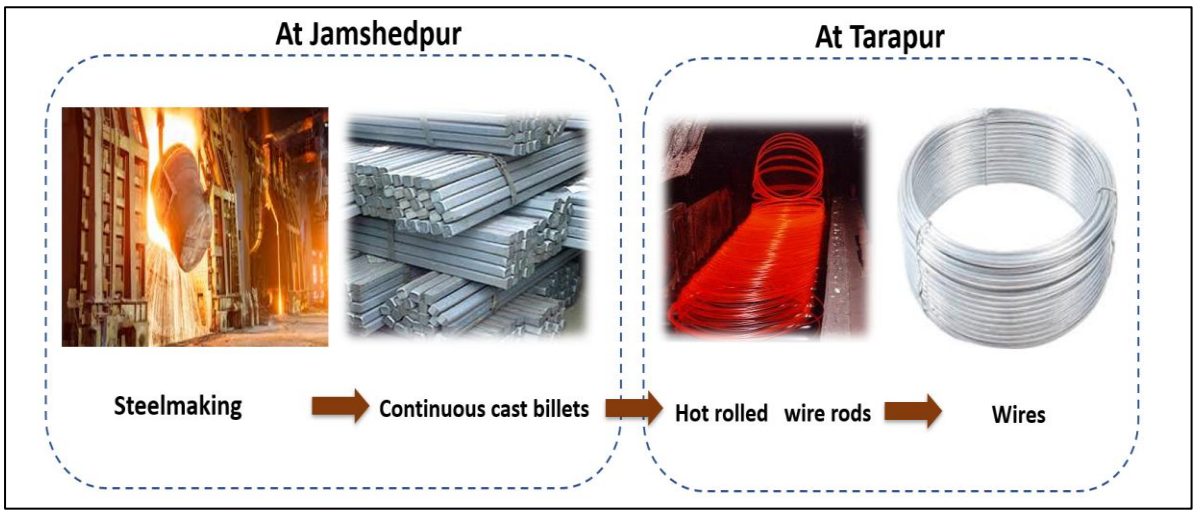

Figure: Process overview for Steel billet manufacturing

Product Application

Low Relaxed Pre-stressed concrete (LRPC) is used in construction of prestressed concrete structures of various types. It is placed in the concrete structure, elastically stretched and locked with the structure in such a way that when applied force is removed, residual compressive stress in the strand gets transferred to the structure. Thus, the concrete member gets 'pre-stressed' in compression. Structural design for load bearing capacity decides diameter of strand and number of pieces to be used.

Product Range (Size and Grade details)

Manufacturing Unit	Product	Size	Grades
Tarapur (Maharashtra), Pithampur (Madhya Pradesh)	Low Relaxed Pre-stressed concrete (LRPC) wire	9.5 to 15.7 mm	1770, 1860, 1960 MPa

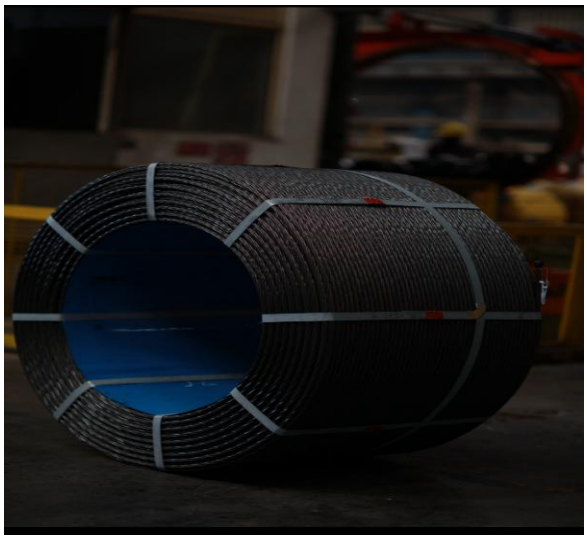

Figure: Illustrative Low Relaxed Pre-stressed concrete (LRPC) wire

Content Declaration

The mass (weight) of one unit of a product, as purchased or per declared unit:	1 Tonne (1000 kg) Tata Steel Low Relaxed Pre-stressed concrete (LRPC) wire
The mass and the content of distribution and/or consumer packaging	The packaging material is having minimal impacts, so it has not been considered in the assessment
Other information on substances with hazardous and toxic properties	Not Applicable

Content of the product in the form of a list of materials and substances, and their mass

Product components	Mass, kg	Post-consumer material, Weight (%)	Biogenic material, weight % of product	Biogenic material, kg C/product or declared unit
Steel (0.27C-Fe)	1000	4	0	0
Iron	997.3	-	0	0
Carbon	2.7	-	0	0
Total	1000	-	0	0

1 kg biogenic carbon in the product/packaging is equivalent to the uptake of 44/12 kg of CO_2. As the packaging material has minimal impact, it was excluded from the assessment and its biogenic content is therefore not listed in the table. Products do not contain any substances that can be included in "Candidate List of Substances of Very High Concern for Authorization" and raw materials used are not part of the EU REACH regulation.

Life Cycle Assessment

LCA information	
Functional unit / declared unit	**1 Tonne (1000 kg) Tata Steel Low Relaxed Pre-stressed concrete (LRPC)**
Conversion factor to mass	1
Reference service life	Not applicable
Time representativeness	The collection of foreground data refers to April 2022 to March 2023
Geographical scope	India (A1-A3, C1-C4, D)
Database(s) and LCA software used	The LCA model was created using Sphera's LCA for Experts (LCA FE) software, version 10.9
Description of system boundaries	Cradle to gate with modules C1-C4 and module D (A1-A3 + C + D)
Reference package used	EN 15804 reference package based on EF 3.1

System Diagram

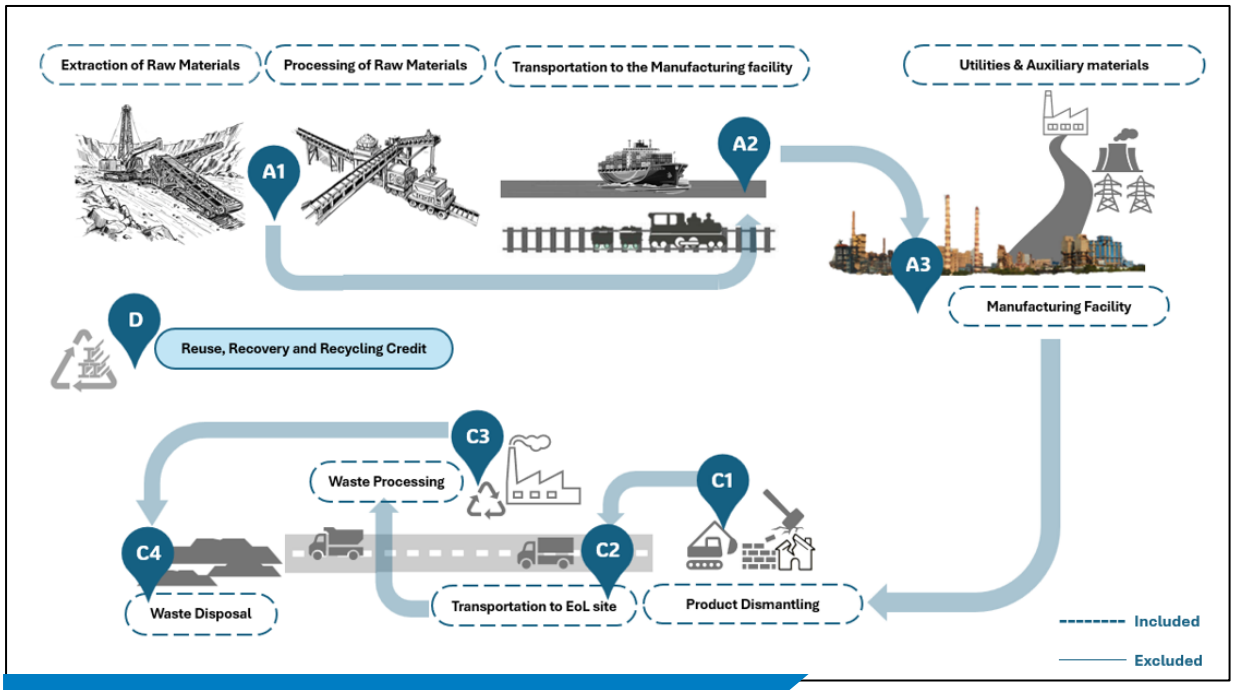

Figure: System boundary diagram

Module A1 – A3

The product stage includes provision of all materials, products and energy, as well as impacts of raw material transportation and intermediate products, waste processing up to the end-of waste state or disposal of final residues during the product stage. The transport distances and routes are calculated based on the manufacturer data. The electricity has been sourced from electricity grid and the emission factor of the electricity mix for the GWP-GHG indicator is 1.02 kg CO_2eq./kWh (Sphera MLC 2024.2). The electricity grid is powered by a mix of energy sources, with hard coal contributing the largest share (59%), followed by lignite (12%), hydroelectric power (11%), natural gas (4%), wind energy (5%), solar photovoltaics (4%), nuclear energy (3%) and others (2%). The remaining portion comes from coal gases, fuel oil, waste, and biogas. There is no packaging considered in the present assessment.

Module C1 – C4

Within this EPD, the modules C1-C4 are included. These modules consider the dismantling of the considered product (C1), the transportation of the dismantled components to their End of Life (EoL) destination (C2), the waste processing for recovery or recycling (C3) as well as the disposal (C4), if given. At EoL, the steel material leaves the product system in C3 for recycling in Module D. The considered EoL scenario for the steel material is 88% recycling and 12% landfill.

After use the product is transported to waste processing. In the C module, default values provided by the PCR 2019:14 v.2.0.1 were used for demolition/deconstruction (C1) as no specific data was obtained. The default values for transport distances to waste treatment (C2) were also used, 80 km for materials not to be incinerated and 130 km for materials to be incinerated.

Module D

Module D includes any declared benefits and loads from net flows leaving the product system that have not been allocated as co-products and that have passed the end-of-waste state in the form of reuse, recovery and/or recycling potentials. Metals are assumed to reach the end of waste status directly at the construction site. The treatment as well as net benefits and loads of reuse or recycling potentials (for the net scrap amount only) are grouped to module D. In the EoL net scrap approach (Amount of steel recycled at EoL – Scrap input from previous product life cycles) has been used. The environmental burden for the extra scrap used in the manufacturing has been considered in module D.

Cut-off Criteria

The environmental impact of the product studied has been assessed by considering all significant processes, materials, and emissions. Excluded flows are assumed to have a negligible impact, each excluded material flow does not exceed 1% of mass, energy or environmental relevance for each unit process and contributing less than 5% to the cumulative impact assessment categories. The production of capital equipment, facilities, and infrastructure required for manufacture has not been considered.

Data quality and sources

Data quality is compliant with ISO 14025:2006. All primary data were collected for the financial year 2023. All background data come from the Sphera MLC 2024.2.

Allocation

With any multi-product system, allocation rules are defined to relate the system inputs and outputs to each of the products. Several methods are documented in ISO 14040:2006 and ISO Technical Report 14049.

Modules declared, geographical scope, share of specific data (in GWP-GHG results) and data variation (in GWP-GHG results)

Modules of the production life cycle included (X- Module declared, ND- Not declared)

	Product Phase			Construction process stage		Use stage							End of Life stage				Resource recovery stage
	Raw material supply	Transport	Manufacturing	Transport	Construction Installation	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste processing	Disposal	Reuse-Recovery-Recycling-potential
Module	A1	A2	A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
Declared	X	X	X	ND	ND	ND	ND	ND	ND	ND	ND	ND	X	X	X	X	X
Geography	India												India				India
Specific data used	87%																
Variation – products	0%																
Variation – sites	3%																

Data quality assessment and declaration

Description of data quality assessment and reference year	Primary data collection concerns the year 2023 and represent the whole annual production. The quality of the relevant data used for the EPD in terms of time, geography and technology representativeness using EN15804:2012+A2:2019. The relevant data assessed included no “fair” or “poor” data.

Process name	Source type	Source	Reference year	Data category	Share of primary data, of GWP-GHG results for A1-A3
Manufacturing of product	Collected data	EPD owner	FY 2024	Primary data	77.2%
Generation of electricity used in manufacturing of product	Collected data and database	Sphera MLC 2024.2 and EPD owner	2024	Primary data	8.53%
Transport of raw materials	Database	Sphera MLC 2024.2	2024	Primary data	1.53%
Total share of primary data, of GWP-GHG results for A1-A3					87.2%

Results of the environmental performance indicators

Note: The estimated impact results are only relative statements, which do not indicate the endpoints of the impact categories, exceeding threshold values, safety margins and/or risks. If the EPD covers the end-of-life stage: "The results of the end-of-life stage (modules C1-C4) should be considered when using the results of the product stage (modules A1-A3)." For services, "A1-A3" shall be replaced by "A1-A5"

Mandatory impact category indicators according to EN15804+A2:2019

Results per 1 Tonne of Low Relaxation Pre-stressed Concrete (LRPC) wire							
Indicator	Unit	A1 - A3	C1	C2	C3	C4	D
GWP-total	kg CO_2 eq.	3.14E+03	7.26E-02	8.57E+00	2.35E+00	3.57E+00	-1.45E+03
GWP-fossil	kg CO_2 eq.	3.14E+03	7.28E-02	8.58E+00	2.35E+00	3.57E+00	-1.45E+03
GWP-biogenic	kg CO_2 eq.	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
GWP-luluc	kg CO_2 eq.	8.85E-01	8.15E-06	1.94E-04	2.76E-03	1.45E-02	-1.94E-01
ODP	kg CFC 11 eq.	4.96E-09	6.83E-15	1.63E-13	8.99E-12	1.04E-11	-1.96E-09
AP	mol H^+ eq.	1.19E+01	2.21E-04	6.62E-02	3.02E-02	2.08E-02	-3.57E+00
EP-freshwater	kg P eq.	5.46E-04	5.49E-08	1.31E-06	1.68E-06	6.74E-06	-3.4E-04
EP-marine	kg N eq.	1.82E+00	4.42E-05	3.30E-02	3.44E-03	5.03E-03	-5.73E-01
EP-terrestrial	mol N eq.	2.00E+01	4.87E-04	3.62E-01	3.77E-02	5.53E-02	-4.89E+00
POCP	kg NMVOC eq.	5.49E+00	2.38E-04	6.37E-02	1.98E-03	1.58E-02	-2.22E+00
ADP-minerals & metals*	kg Sb eq.	2.24E-04	4.23E-09	6.4E-08	3.51E-08	2.23E-07	-7.87E-03
ADP-fossil*	MJ	2.72E+04	4.70E+00	7.12E+01	3.95E+01	5.53E+01	-1.38E+04
WDP*	m^3	2.29E+02	4.41E-04	6.56E-03	3.63E-01	4.08E-01	-9.38E+01
Acronyms	GWP - total = global warming potential; GWP - fossil = global warming potential (fossil fuel only); GWP - biogenic = global warming potential (biogenic); GWP - luluc = global warming potential (land use only); ODP = ozone depletion; AP = acidification terrestrial and freshwater; EP freshwater = eutrophication potential (freshwater); EP - marine = eutrophication potential (marine); EP- terrestric = eutrophication potential (terrestrial); POCP = photochemical ozone formation; ADPE = abiotic depletion potential (element); ADPF = abiotic depletion potential (fossil); WDP = water scarcity						

Additional mandatory and voluntary impact category indicators

Results per 1 Tonne of Low Relaxation Pre-stressed Concrete (LRPC) wire							
Indicator	Unit	A1-A3	C1	C2	C3	C4	D
GWP-GHG*	kg CO_2 eq.	3.14E+03	7.26E-02	8.57E+00	2.35E+00	3.57E+00	-1.38E+03
Acronyms	GWP - GHG*= global warming potential (greenhouse gases) This indicator includes all greenhouse gases included in GWP-total but excludes biogenic carbon dioxide emissions and uptake and biogenic carbon stored in the product with characterization factors (CFs) based on IPCC (2013)						

Resource use indicators according to EN15804+A2:2019

Results per 1 Tonne of Low Relaxation Pre-stressed Concrete (LRPC) wire							
Indicator	Unit	A1-A3	C1	C2	C3	C4	D
PERE	MJ	1.87E+03	9.70E-03	2.31E-01	5.87E+00	8.20E+00	-5.73E+02
PERM	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PERT	MJ	1.87E+03	9.70E-03	2.31E-01	5.87E+00	8.20E+00	-5.73E+02
PENRM	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PENRE	MJ	2.72E+04	4.70E+00	1.12E+02	3.12E+01	5.53E+01	-1.45E+04
PENRT	MJ	2.72E+04	4.70E+00	1.12E+02	3.12E+01	5.53E+01	-1.45E+04
SM	kg	4.00E+01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
RSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NRSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
FW	M^3	6.15E+00	1.36E-05	3.23E-04	9.09E-03	1.23E-02	-2.0E+00
Acronyms	PERE = Use of renewable primary energy excluding renewable primary energy resources used as raw materials; PERM = Use of renewable primary energy resources used as raw materials; PERT = Total use of renewable primary energy resources; PENRE = Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials; PENRM = Use of non-renewable primary energy resources used as raw materials; PENRT = Total use of non-renewable primary energy re-sources; SM= Use of secondary materials RSF = Use of renewable secondary fuels; NRSF = Use of non-renewable secondary fuels; FW = Use of net fresh water						

Output flows & Waste indicators according to EN15804+A2:2019

Results per 1 Tonne of Low Relaxation Pre-stressed Concrete (LRPC) wire							
Indicator	Unit	A1-A3	C1	C2	C3	C4	D
HWD	kg	1.97E-05	5.98E-11	1.42E-09	9.25E-09	1.32E-08	-1.01E-05
NHWD	kg	3.71E+00	7.20E-05	1.72E-03	9.22E-03	1.60E+02	-1.52E+00
RWD	kg	9.67E-02	5.89E-07	1.40E-05	3.12E-04	7.50E-04	-1.51E-03
MFR	kg	0.00E+00	0.00E+00	0.00E+00	8.80E+02	0.00E+00	0.00E+00
MER	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
EEE	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
EET	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Acronyms	HWD = Hazardous waste disposed; NHWD = Non-hazardous waste disposed; RWD = Radioactive waste disposed; MFR = Materials for recycling; MER = Materials for energy recovery; EEE = Exported electrical energy; EET = Exported thermal energy						

Additional environmental performance indicators according to EN15804+A2:2019

Results per 1 Tonne of Low Relaxation Pre-stressed Concrete (LRPC) wire							
Indicator	Unit	A1-A3	C1	C2	C3	C4	D
PM	Disease incidences	1.78E-04	2.91E-09	4.81E-07	4.95E-07	2.40E-07	-3.35E-05
IR	kBq U235 eq.	9.14E+00	5.33E-05	1.27E-03	2.89E-02	1.03E-01	0.00E+0
ETF	CTUe	7.31E+03	2.12E+00	5.05E+01	2.26E+01	3.36E+01	-7.55E+01
HTP-c	CTUh	1.34E-07	3.35E-11	8.08E-10	3.71E-10	7.32E-10	-4.46E-08
HTP-nc	CTUh	5.23E-06	6.97E-10	1.68E-08	1.73E-08	2.60E-08	-1.53E-06
SQP	Pt	1.94E+03	5.61E-03	1.34E-01	4.47E+00	1.01E+01	-3.43E+02
Acronyms	PM = Particulate matter emissions; IR = Ionising radiation, human health; ETF= Eco-toxicity (freshwater); HTP-c = Human toxicity, cancer effects; HTP-nc = Human toxicity, non-cancer effects; SQP = Soil quality potential/Land use related impacts						

Disclaimer: The results of this environmental impact indicator shall be used with care as the uncertainties of these results are high or as there is limited experience with the indicator.

Additional environmental information

End of Life scenarios

The following tables show the environmental impact of the end-of-life stages, assuming that 100% of 1 tonne of product is recycled.

Indicator	Unit	C1	C2	C3	C4	D
GWP-total	kg CO_2 eq.	7.20E-02	5.36E+00	2.80E+00	0.00E+00	-1.73E+03
GWP-fossil	kg CO_2 eq.	7.23E-02	5.36E+00	2.80E+00	0.00E+00	-1.73E+03
GWP-biogenic	kg CO_2 eq.	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
GWP-lulc	kg CO_2 eq.	8.09E-06	1.21E-04	3.29E-03	0.00E+00	-2.31E-01
ODP	kg CFC 11 eq.	6.77E-15	1.02E-13	1.07E-11	0.00E+00	-2.33E-09
AP	mol H^+ eq.	2.19E-04	4.14E-02	3.60E-02	0.00E+00	-4.25E+00
EP-freshwater	kg P eq.	5.45E-08	8.18E-07	2.00E-06	0.00E+00	-4.05E-04
EP-marine	kg N eq.	4.39E-05	2.06E-02	4.10E-03	0.00E+00	-6.82E-01
EP-terrestrial	mol N eq.	4.83E-04	2.26E-01	4.49E-02	0.00E+00	-5.82E+00
POCP	kg NMVOC eq.	2.37E-04	3.98E-02	2.36E-03	0.00E+00	-2.64E+00
ADP-minerals & metals*	kg Sb eq.	4.20E-09	6.30E-08	4.18E-08	0.00E+00	-9.37E-03
ADP-fossil*	MJ	4.67E+00	7.00E+01	4.70E+01	0.00E+00	-1.64E+04
WDP*	m^3	4.38E-04	6.56E-03	4.32E-01	0.00E+00	-1.12E+02
Acronyms	GWP - total = global warming potential; GWP - fossil = global warming potential (fossil fuel only); GWP - biogenic = global warming potential (biogenic); GWP - luluc = global warming potential (land use only); ODP = ozone depletion; AP = acidification terrestrial and freshwater; EP freshwater = eutrophication potential (freshwater); EP - marine = eutrophication potential (marine); EP- terrestric = eutrophication potential (terrestrial); POCP = photochemical ozone formation; ADPE = abiotic depletion potential (element); ADPF = abiotic depletion potential (fossil); WDP = water scarcity					

The following tables show the environmental impact of the end-of-life stages, assuming that 100% of 1 tonne of product is sent to landfill.

Indicator	Unit	C1	C2	C3	C4	D
GWP-total	kg CO_2 eq.	7.20E-02	5.36E+00	0.00E+00	2.23E+01	0.00E+00
GWP-fossil	kg CO_2 eq.	7.23E-02	5.36E+00	0.00E+00	2.23E+01	0.00E+00
GWP-biogenic	kg CO_2 eq.	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
GWP-lulc	kg CO_2 eq.	8.09E-06	1.21E-04	0.00E+00	9.06E-02	0.00E+00
ODP	kg CFC 11 eq.	6.77E-15	1.02E-13	0.00E+00	6.50E-11	0.00E+00
AP	mol H^+ eq.	2.19E-04	4.14E-02	0.00E+00	1.30E-01	0.00E+00
EP-freshwater	kg P eq.	5.45E-08	8.18E-07	0.00E+00	4.21E-05	0.00E+00
EP-marine	kg N eq.	4.39E-05	2.06E-02	0.00E+00	3.14E-02	0.00E+00
EP-terrestrial	mol N eq.	4.83E-04	2.26E-01	0.00E+00	3.46E-01	0.00E+00
POCP	kg NMVOC eq.	2.37E-04	3.98E-02	0.00E+00	9.88E-02	0.00E+00
ADP-minerals & metals*	kg Sb eq.	4.20E-09	6.30E-08	0.00E+00	1.39E-06	0.00E+00
ADP-fossil*	MJ	4.67E+00	7.00E+01	0.00E+00	3.46E+02	0.00E+00
WDP*	m^3	4.38E-04	6.56E-03	0.00E+00	2.50E+00	0.00E+00
Acronyms	GWP - total = global warming potential; GWP - fossil = global warming potential (fossil fuel only); GWP - biogenic = global warming potential (biogenic); GWP - luluc = global warming potential (land use only); ODP = ozone depletion; AP =					

	acidification terrestrial and freshwater; EP freshwater = eutrophication potential (freshwater); EP - marine = eutrophication potential (marine); EP- terrestric = eutrophication potential (terrestrial); POCP = photochemical ozone formation; ADPE = abiotic depletion potential (element); ADPF = abiotic depletion potential (fossil); WDP = water scarcity

Abbreviation

Abbreviation	Definition	Abbreviation	Definition
General Abbreviations			
EN	European Norm (Standard)	**ISO**	International Organization for Standardization
EPD	Environmental Product Declaration	**LCA**	Life Cycle Assessment
EF	Environmental Footprint	**PCR**	Product Category Rules
GPI	General Programme Instructions	**c-PCR**	Complementary Product Category Rules
BOF	Basic Oxygen Furnace	**LCI**	Life Cycle Inventories
MTPA	Million Tonnes Per Annum	**CEN**	European Committee for Standardization
Environmental Impact Indicators (EN 15804)			
GHG	Greenhouse gas	**EP**	Eutrophication Potential
GWP	Global Warming Potential (kg CO_2 eq.)	**EP-freshwater**	Freshwater eutrophication potential (kg P eq.)
GWP-fossil	Global Warming Potential from fossil sources (kg CO_2 eq.)	**EP-marine**	Marine eutrophication potential (kg N eq.)
GWP-biogenic	Global Warming Potential from biogenic sources (kg CO_2 eq.)	**EP-terrestrial**	Terrestrial eutrophication potential (mol N eq.)
GWP-luluc	Global Warming Potential from land use and land use change (kg CO_2 eq.)	**POCP**	Photochemical Ozone Creation Potential (kg NMVOC eq.)
GWP-total	Total Global Warming Potential (kg CO_2 eq.)	**ADP**	Abiotic Depletion Potential
GWP-GHG	Global Warming Potential for greenhouse gases (kg CO_2 eq.)	**ADP-minerals & metals**	Abiotic depletion potential for non-fossil resources (kg Sb eq.)
ODP	Ozone Depletion Potential (kg CFC-11 eq.)	**ADP-fossil**	Abiotic depletion potential for fossil resources (MJ)
AP	Acidification Potential (mol H^+ eq.)	**WDP**	Water Deprivation Potential (m^3)
Resource Use Indicators			
PERE	Use of renewable primary energy excluding renewable primary energy resources used as raw materials (MJ)	**PENRT**	Total use of non-renewable primary energy resources (MJ)
PERM	Use of renewable primary energy resources used as raw materials (MJ)	**SM**	Use of secondary material (kg)
PERT	Total use of renewable primary energy resources (MJ)	**RSF**	Use of renewable secondary fuels (MJ)
PENRE	Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials (MJ)	**NRSF**	Use of non-renewable secondary fuels (MJ)
PENRM	Use of non-renewable primary energy resources used as raw materials (MJ)	**FW**	Use of net fresh water (m^3)
Waste and Output flow indicators			
HW	Hazardous Waste (disposed) (kg)	**MR**	Material for Recycling (kg)
NHW	Non-Hazardous Waste (disposed) (kg)	**MER**	Materials for Energy Recovery (kg)
RW	Radioactive Waste (disposed) (kg)	**EEE**	Exported Energy, Electricity (MJ)
CFR	Components for Reuse (kg)	**EET**	Exported Energy, Thermal (MJ)

Reference

- General Programme Instructions of the International EPD® System. Version 5.0.1
- PCR 2019:14. Construction Products, Version 2.0.1 [valid upto: 2030-04-07]
- ISO 14025: EN ISO 14025:2011-10 Environmental labels and declarations - Type III environmental declarations - Principles and procedures
- ISO 14040: EN ISO 14040:2009-11 Environmental management - Life cycle assessment - Principles and framework
- LCA FE: LCA FE Software System and Database for Life Cycle Engineering, Sphera Solution GmbH, Leinfelden-Echterdingen, 2024 (https://sphera.com/solutions/product-stewardship/life-cycle-assessment-software-and-data/managed-lca-content/)
- WSI and Eurofer's Co-Product Allocation Methodology 2014 – A methodology to determine the LCI of Steel Industry Co-products
- World Steel Association- CO2 Data Collection User Guide, Version 9 (May 2019).
- ISO 14044: EN ISO 14044:2006-10 Environmental management - Life cycle assessment - Requirements and guidelines

Version History

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